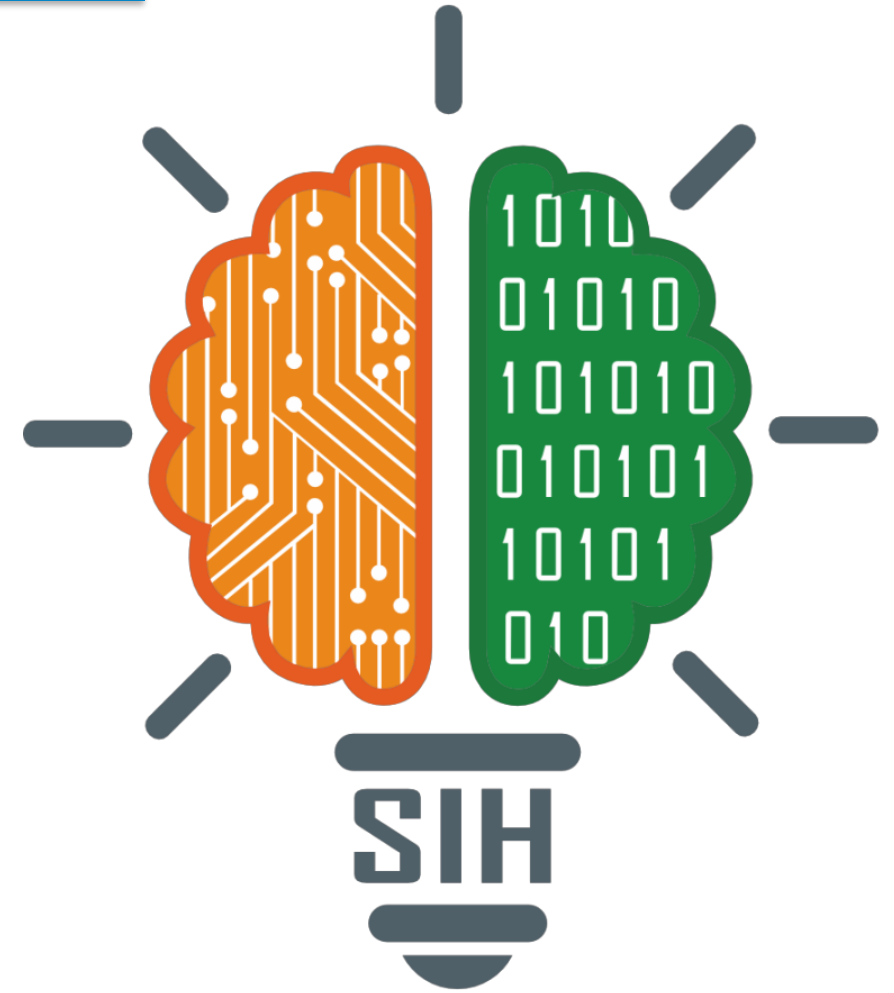


Adaptive Foveated 2.5D Semantic Mapping for Autonomous Navigation

- **Problem Statement ID** – 26053

- **Problem Statement Title** – Adaptive Foveated 2.5D Semantic Mapping from LiDAR Point Clouds for Autonomous Navigation
- **Theme** – Smart Vehicles
- **PS Category** – Software
- **Team ID** – TBA
- **Team Name** – TeraPulse
- **Team Members** – Abhay Verma (Team Leader), Lokendra Singh, Yashvardhan Jain, Chetan Meena, Sachin Chaubey, Esha Verma
- **Institute** – Indian Institute of Information Technology Bhopal



Adaptive Foveated 2.5D Semantic Mapping for Autonomous Navigation

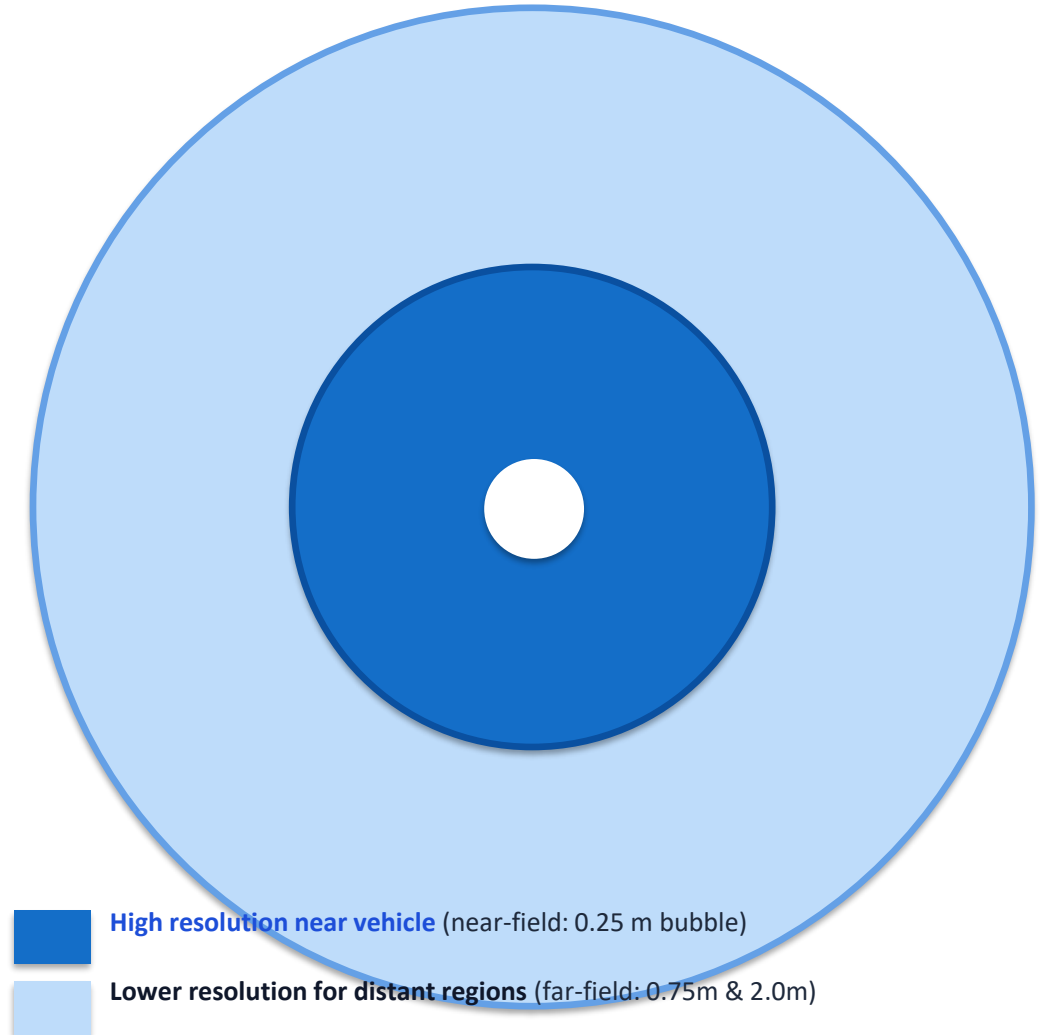
❖ Proposed Solution

- **2.5D Semantic Elevation Map:** Generates an efficient multi-resolution heightmap from raw 3D LiDAR point clouds.
- **High-Precision Near Field (0–10 m):** Fine 0.25 m cells for sub-meter obstacle clearance, terrain slope, and reactive steering.
- **Bandwidth-Saving Far Field:** Gradually reduces to 0.75 m & 2.0 m resolution to slash unnecessary distant computations.
- **Full Elevation Profiling:** Preserves height, step slope & obstacle boundaries without full 3D voxel memory overload.
- **Real-Time Terrain Intelligence:** Accurately classifies drivable ground, static obstacles, and dynamic moving objects.
- **Autonomous Navigation Stack:** Enables instant A* heuristic path planning and smooth Pure Pursuit vehicle steering.

❖ Innovation & Uniqueness

Foveated adaptive resolution focuses computational resources where they matter most — slashing memory by **66.2%** and cutting latency to **24.23 ms (>35 FPS)**.

Foveated 2.5D Semantic Elevation Map



System Workflow Pipeline

[1] LiDAR Point Cloud Ingestion

[2] Preprocessing & Noise Removal

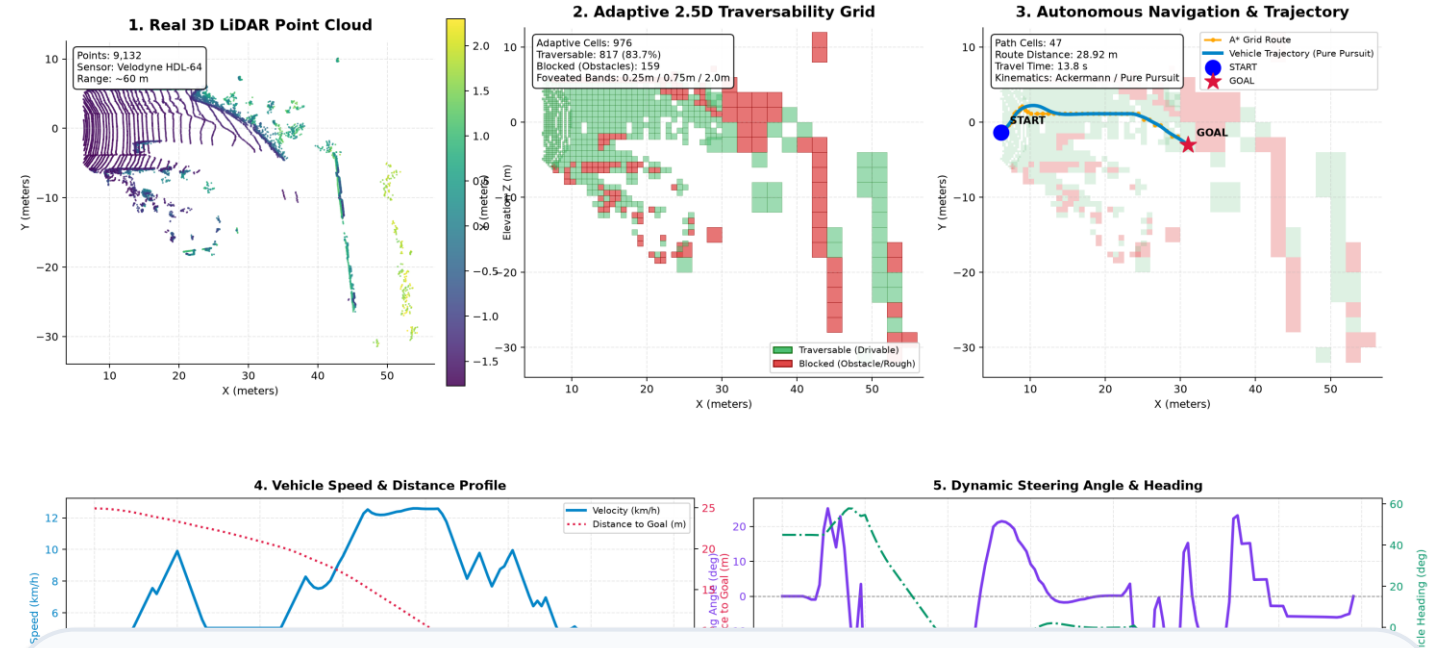
[3] Deep Semantic Classification

[4] Foveated Adaptive Grid Generation

[5] 2.5D Semantic Elevation Map

[6] Real-Time Visualization & Navigation

TERA PULSE — SIH26053 Autonomous Navigation System Adaptive Variable-Resolution 2.5D LiDAR Mapping & Dynamic Control



Technologies & Software Implementation:

- **Core Stack:** Python 3.12, NumPy (vectorized hashing), Open3D (point cloud geometry)
- **Semantic Intelligence:** 4-Class ML Classifier (Ground, Static Obstacles, Dynamic Vehicles)
- **Autonomous Motion:** A* Graph Search, Catmull-Rom Splines, Ackermann Pure Pursuit Kinematics
- **Live Web Dashboard:** Flask REST API, 60 FPS HTML5 Canvas with dynamic moving-car foveation

✓ Feasibility & Viability

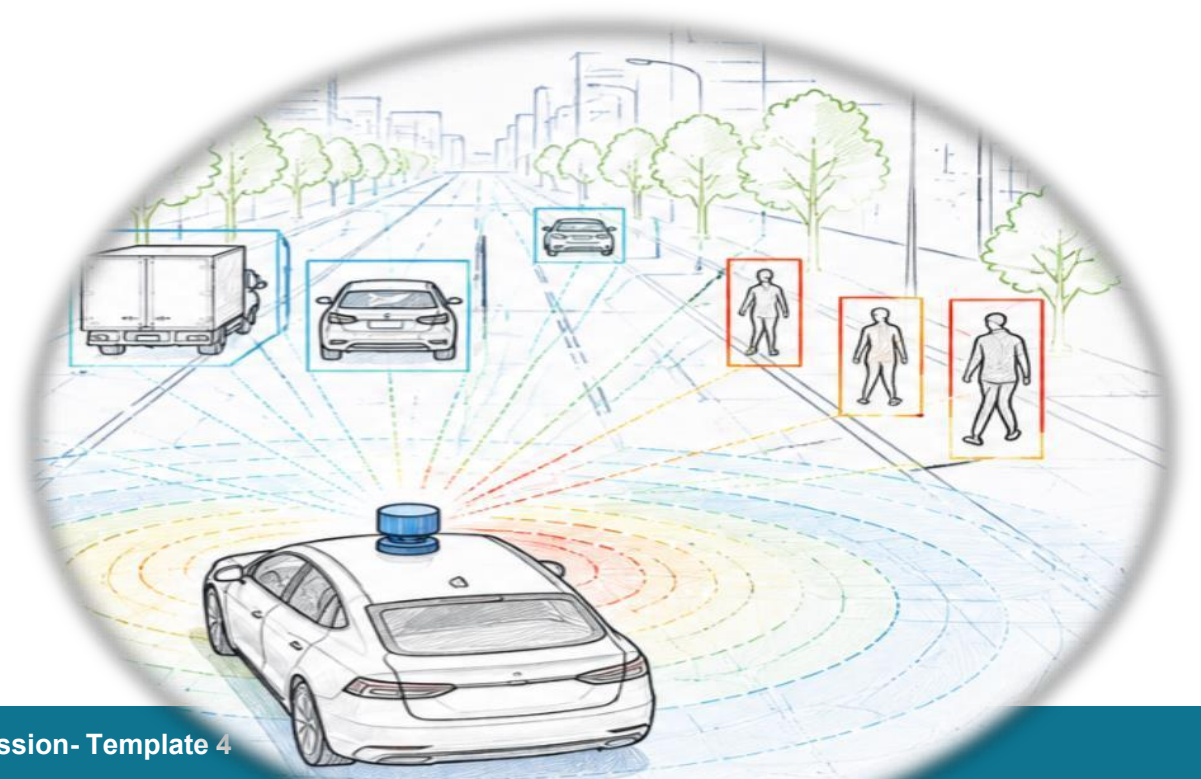
- **Open-Source & Standard Datasets:** Built on widely tested LiDAR datasets and open tools.
- **Rapid Edge Prototype:** Python 3.12, Open3D, and PyTorch enable agile modular development.
- **Edge Compute Efficiency:** Adaptive resolution drastically reduces RAM and latency.
- **Independent Subsystem Verification:** Modular pipeline with 5/5 automated unit tests passing.

⚠ Tactical Challenges

- **Heavy LiDAR Data Deluge:** Real-time processing of 1.5M points/sec from 10–20 Hz sweeps.
- **Unstructured Off-Road Terrain:** Accurate terrain segmentation in dusty, rocky environments.
- **Bridge / Underpass False Obstacles:** Preventing ceiling points from falsely blocking tunnels.
- **Preserving Geometry in Low-Res:** Retaining safety clearance while reducing distant cell detail.

🛡 Mitigation Strategies

- **Foveated Attention Hashing:** Focuses 0.25 m resolution in near-field for instant 24.23 ms execution.
- **SOR & RANSAC Ground Fit:** Statistical filters eliminate dust scatter and extract clean ground plane.
- **Vertical Clearance Filtering:** Clips points >2.2 m to allow safe tunnel traversal without 3D voxels.
- **Rigorous Verification:** Benchmarked on 20,672 points at >35 FPS with zero crash failures.



🔧 Strategic Defence Impact

- **Autonomous Tactical UGVs:** Border patrol (Ladakh/Siachen) & minefield reconnaissance without soldier casualties.
- **Zero-Visibility Warfare:** LiDAR penetrates smoke, sandstorms, fog & darkness where cameras fail.
- **Sub-Meter Hazard Detection:** Guarantees obstacle clearance in safety-critical vehicle zones.
- **Real-Time Terrain Intelligence:** Prevents roll-overs on steep inclines and hidden ditches.

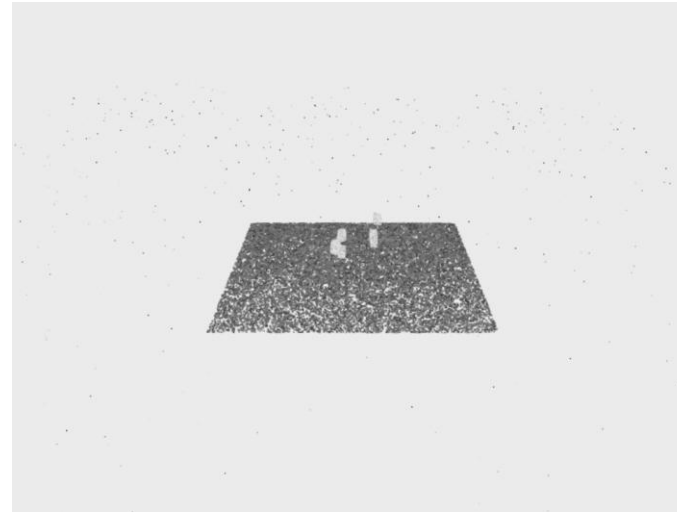
🌐 Dual-Use Applications

- **Defence & Security:** UGVs, logistics convoys & border patrol rovers.
- **Disaster Response:** Search-and-rescue rovers in earthquake rubble & caves.
- **Industrial Autonomy:** Autonomous tractors on farmland & mining dump trucks.

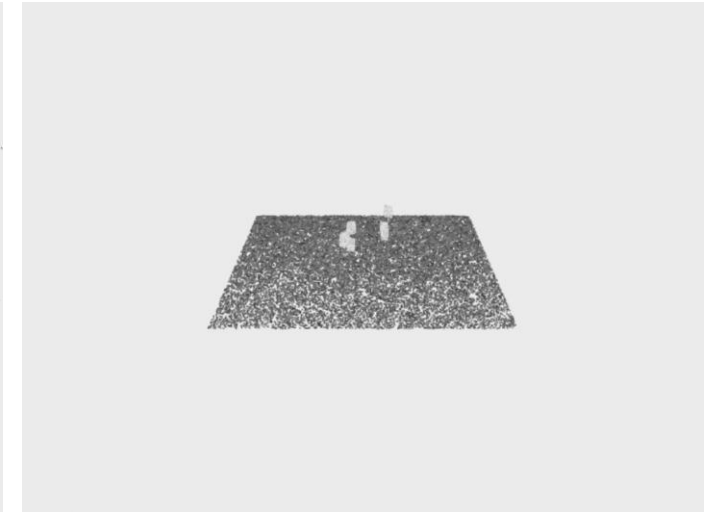
⚡ Measurable Benefits

- **66.2% Cell Reduction:** Slashing memory and compute load on embedded edge hardware.
- **4.8× Faster Latency:** Down from 115.94 ms to 24.23 ms for true 35+ FPS control.
- **Full 2.5D Elevation:** Preserves slopes, roughness & step heights without 3D voxels.

📷 Visual Impact: Raw Point Cloud vs Cleaned 2.5D Elevation Map



Raw LiDAR Scan (20,672 points with noise)



Filtered 2.5D Elevation Map (Drivable Ground Extracted)

🏆 Production Performance Benchmark (KITTI Velodyne HDL-64E):

- **Active Stored Cells:** Reduced from 2,888 (uniform) to 976 (adaptive) — **66.2% Cell Reduction**
- **Mapping Compute Time:** Reduced from 115.94 ms to 24.23 ms — **4.8× Latency Speedup**
- **Traversable Corridor:** 817 safe cells (83.71%) accurately identified without collision risk
- **Autonomous Trajectory:** 47-cell A* path tracked smoothly via Pure Pursuit (10 Hz Telemetry)

 Research Areas

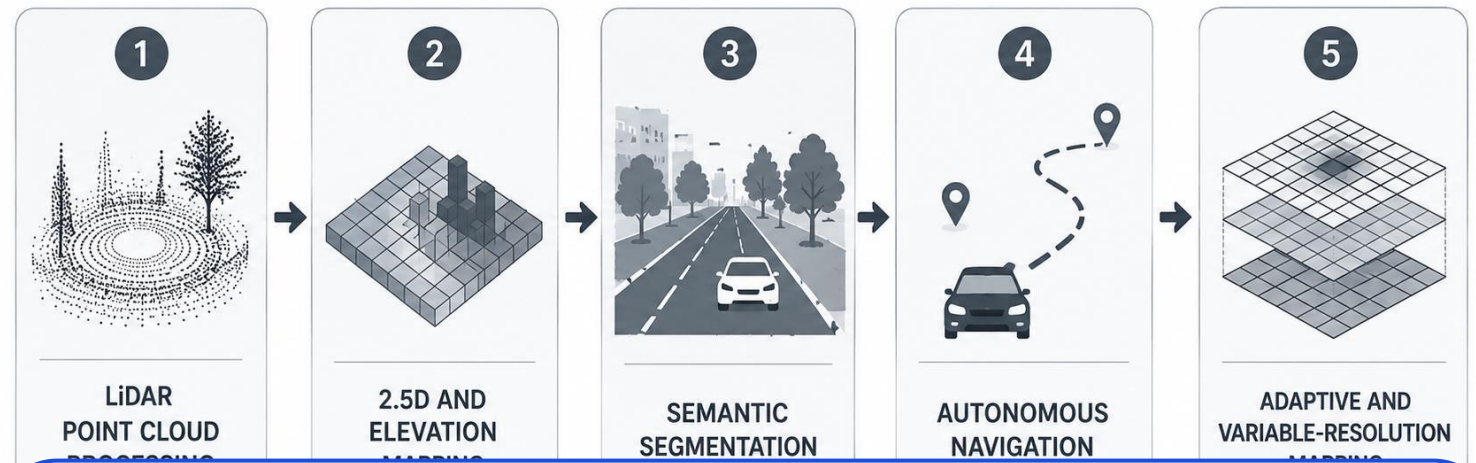
- **LiDAR Processing:** Point cloud filtering, SOR & RANSAC ground estimation.
- **2.5D Elevation Mapping:** Robot-centric height & slope representation.
- **Semantic Understanding:** Deep learning segmentation of large-scale point clouds.
- **Autonomous Navigation:** A* heuristic planning & Pure Pursuit vehicle kinematics.
- **Adaptive Spatial Hashing:** Foveated variable-resolution concentric indexing.

 Academic References

- **KITTI Vision Suite:** Geiger et al. (Karlsruhe / Toyota Tech), Velodyne HDL-64E LiDAR benchmark.
- **ETH Zurich Elevation Mapping:** P. Fankhauser et al. (IEEE IROS), Robot-Centric Elevation Mapping.
- **CMU Pure Pursuit Tracker:** R. C. Coulter (Carnegie Mellon Robotics Institute), Pure Pursuit Kinematics.
- **Large-Scale Segmentation:** Q. Hu et al. (Oxford / CVPR), RandLA-Net Point Cloud Segmentation.
- **Foveated Visual Attention:** J. Martinez et al. (IEEE Trans. Robotics), Foveated Saliency for Navigation.

LiDAR 2.5 RESEARCH PIPELINE

LiDAR 2.5 RESEARCH PIPELINE

 Verified Project Deliverables & Repository:

- ✓ **Interactive Localhost Simulator:** 60 FPS HTML5 / Flask Dashboard with live dynamic foveation & HUD
- ✓ **Automated Test Suite:** 5/5 comprehensive unit tests (tests/test_pipeline.py) passing in 4.0s
- ✓ **Open-Source Codebase:** Production GitHub repository: github.com/AbhayVerma628/SIH26053_Lidar
- ✓ **Autonomous Output Artifacts:** 5-panel dashboard, 10 Hz telemetry CSV, planned A* path, benchmark JSON